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A PALEOMAGNETIC AND ISOTOPIC AGE FOR THE WACKERFIELD DYKE OF NORTHERN ENGLAND

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The Wackerfield Dyke is exposed in northern England and its field relationships suggest it could be related to the Eocene Mull Dyke Swarm. Wackerfield Dyke specimens taken from below the zone of surface weathering have a natural remanent magnetization that is extremely stable to thermal and alternating field demagnetization. Stability is expected because the Fe–Ti oxides are highly oxidized. The stable magnetization is identical with a primary remanence that can be isolated from certain samples of the Whin Sill of northern England, and gives a mean pole at 175°E, 22°N (dp = 2°, dm = 3°). An Upper Carboniferous age for the Wackerfield Dyke is confirmed by a K–Ar whole rock age of 303 my.

1. Introduction

Basic dykes and sills are common in Northern England and reflect two ages of igneous activity. The older is typified by the Upper Carboniferous Whin Sill, a system of transgressive quartz dolerite sheets exposed across Northumberland and Durham [1, 2]. The younger is represented by the Cleveland–Armathwaite and related dykes (fig. 1), which are an extension of the Eocene Mull Dyke Swarm [3, 4].

The Wackerfield Dyke near Staindrop, County Durham, has been described in some detail [5]. The dyke does not outcrop continuously, but its field relations suggest that it could be part of the Cleveland–Armathwaite Dyke. Nevertheless, the mineralogy of the Wackerfield Dyke shows more affinity with the Whin Sill than with the Tertiary dykes [5]. Paleomagnetic studies have already been made on both the Cleveland–Armathwaite Dyke [6, 7] and the Whin Sill [8, 9], so that a magnetic/isotopic study of the Wackerfield can be directly related to data from these intrusives.

2. Sampling

The Wackerfield Dyke has been exposed by quarrying at 4 places [5]. We drilled 10 samples for thermal demagnetization and a further 8 for alternating field (AF) demagnetization. All orientations were by sun compass. Specimens were cut from either end of each core: they will be referred to as weathered or fresh respectively.

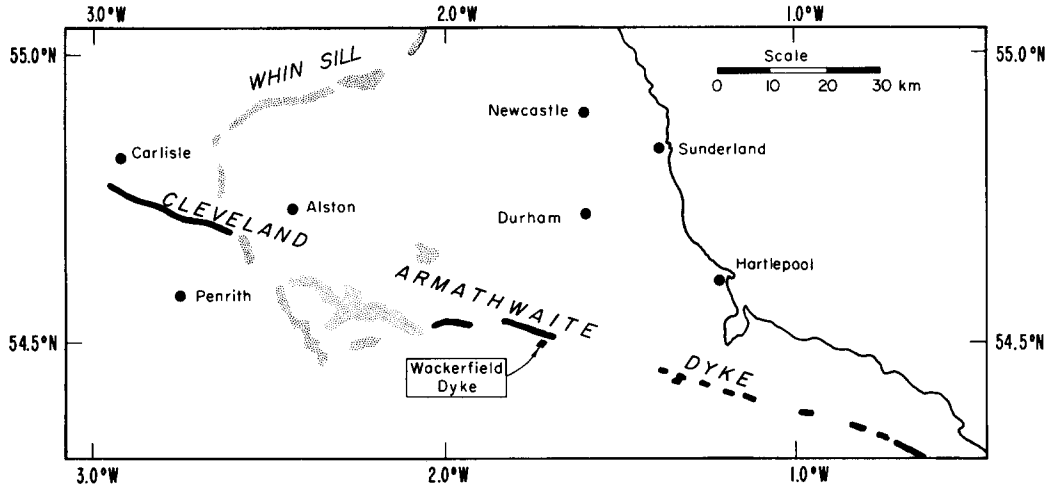


Fig. 1. Map of northern England showing outcrops of the Cleveland–Armathwaite Dyke and the Wackerfield Dyke (solid lines), and the Whin Sill (Stippled): after [5], [7] and [8].

3. Isotopic age

A potassium–argon whole rock age of 303 my was determined for one of the fresh specimens according to methods outlined elsewhere [10]. This corresponds to an Upper Carboniferous age for the intrusion, and thus suggests that the dike is in fact related to the Whin Sill [2].

4. Thermal demagnetization

Sixteen specimens (representing 10 samples) were thermally demagnetized in a shielded space with residual fields less than 5 gamma, and measured with a PAR SM-1 spinner [11]. There are three differences in magnetic behaviour between the weathered (A) specimens and the fresher (B) specimens. First, the initial NRM intensities of the fresh specimens are greater by a factor up to 200. Second, demagnetization of the weathered specimens reveals the presence of significant amounts of secondary magnetization. This is illustrated in fig. 2a, by the demagnetization response for samples 1 and 6. For the fresh specimens from each sample the NRM directions remain virtually unchanged up to 530°C, implying that the remanence is quite stable. The initial NRM directions of the weathered specimens are dispersed, and move toward those of the fresh specimens only at high temperatures. Note in fig. 2a that measurements of the initial NRM for specimen 1A on successive days differ by 35° of arc. Third, the normalized NRM decay curves for the fresh specimens show little change in intensity as far as 530°C (fig. 2b). This again implies stability of the NRM. The weathered specimens show an increase in intensity during thermal demagnitization (e.g. 7A in fig. 2b). This is commensurate with removal of a secondary component of opposite polarity, very probably acquired along the present field.

5. Alternating magnetic field demagnetization

Fresh specimens from the eight separately oriented samples were subjected to incremental AF demagnetization up to 845 peak oersteds and the NRM directions were measured by the rotation beneath a low sensitivity astatic magnetometer [12]. For all the specimens there was no change in the NRM direction until above 300–500 Oe (fig. 2c) and the NRM decay curves were identical, being fairly convex. Both these features confirm that the NRM is stable.

6. Opaque petrography

There is a different mineralogy in the fresh and in the weathered rocks. In the *fresh* specimens, titano-

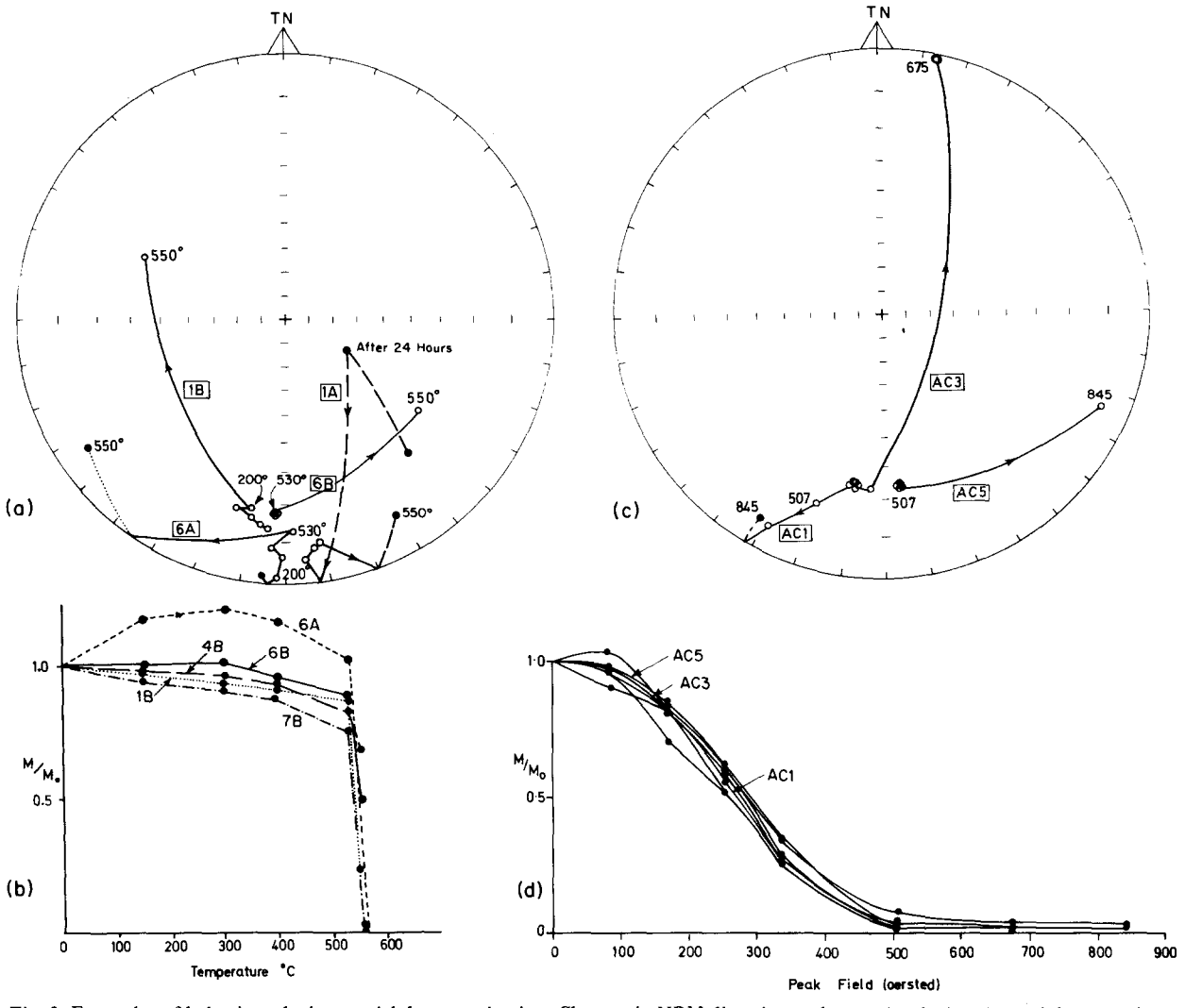


Fig. 2. Examples of behaviour during partial demagnetization. Changes in NRM direction and intensity during thermal demagnetisation are shown in (a) and (b) and during AF demagnetization in (c) and (d). Open and solid circles are, respectively, north seeking magnetization up and down.

magnetite is oxidised along 110 and 111 planes to thick ilmenite lamellae. Spinel rods in the titanomagnetite host place this deuteric oxidation in Class IV to V and infer magnetic stability [13]: this confirms our observation of an NRM very stable to demagnetization. Granulation in the relict titanomagnetite implies hydrothermal alteration [14] and could be related to lenses of pyrite intergrown with the Fe–Ti oxides and sometimes trailing out along dislocations (? perhaps cooling cracks) in the rock. The few discrete ilmenite grains present appear to have been unaffected by the alteration processes.

In the *weathered* specimens, the Fe–Ti oxides have been metasomatically replaced by carbonate–pyrite–anatase–goethite–hematite assemblages, not necessarily always occurring together. In addition, goethite and hematite also occur as rims around the silicate grains and as stringers along cracks in the rock. These observations must account for the very low intensities and secondary magnetization present in such near surface specimens.

The opaque-petrography study raises important questions. In particular, when did the various alteration processes take place, and can they be related to

Table 1
K/Ar whole rock data for the Wackerfield Dyke

K_2O %	Radiogenic argon (mm^3gm^{-1})	% Atmospheric argon	Age (my)
0.958 ± 0.10 (Mean of 3 determinations)	1.05 ± 0.02 (Mean of 2 determinations)	15.0	303 ± 5
$\lambda_e = 0.584 \times 10^{-10}\ yr^{-1}$	$^{40}K/K = 1.19 \times 10^{-2}$ atom %	$\lambda_B = 4.72 \times 10^{-10}\ yr^{-1}$	

Table 2
Paleomagnetic data

	N	Dec	Inc	k	a_{95}
Wackerfield Dyke					
Specimens at 530°C	9	182	− 27	284	3
Specimens at 170 Oe	7	191	− 25	277	4
Overall mean	16	186	− 26	174	3
	(Pole at 175E, 22N, dp = 2; dm = 3)				
Using a stability index [16]					
Most stable (thermal)	9	184	− 26	203	4
Most stable (a.f.)	7	191	− 25	318	3
Whin Sill					
(1) Creer et al. [8] Sites, initial NRM					
* tilt corrected	34	188	− 5	80	3
uncorrected	35	188	− 3	28	5
(2) Storetvedt and Gidskehaug [9] A.f. demag; uncorcorrected					
Specimens	45	193	− 21	178	2
*Sites	5	194	− 22	259	5
Cleveland–Armathwaite Dyke [7]					
Sites, a.f. and thermal demag.	10	160	− 64	144	4

N is number of data in calculation; Dec is declination east of TN; Inc is inclination negative upwards; k is estimate of precision; a_{95} is radius of 95% level confidence [15].

the episodes of metasomatism that affected the Whin Sill [18]. Our own observations suggest that four types of alteration affected the Wackerfield Dyke: (1) deuteric oxidation associated with the time of intrusion, (2) hydrothermal alteration which preferentially affected the relict titanomagnetite, (3) some form of metasomatism which completely destroyed the primary Fe–Ti oxides and (4) Recent surface weathering.

7. Mean directions of magnetization

Mean NRM directions for each demagnetizing stage were calculated according to Fisher [15]. We used data only from those specimens cut from the freshest (i.e. deepest) part of each core, because these showed the greatest magnetic stability. For each demagnetization method 1 sample was rejected from the analysis because the direction was more than $(140/k^{\frac{1}{2}})^\circ$ from the site mean. This meant that there was less than 5% probability that the directions belonged to the population. The NRM of these samples was very stable and showed convex to rectangular NRM decay curves: in all probability they were from dolerite blocks no longer in place.

The mean demagnetization paths are shown in fig. 3, with their associated k values. They confirm that the NRM is extremely stable to both cleaning techniques. There is a small angular discrepancy between the two sets of means, which is mainly in declination. This may be either due to an orientation error (the 2 sets of samples were collected at different times) or more likely the result of using magnetometers in two different labs. Nevertheless an F test [17] shows that the 2 directions are not significantly different at the 95% confidence level.

Listed in table 2 are the mean directions at that temperature (530°C) and A.F. stage (170 Oe) for which there is least scatter i.e. k is highest. In fact they are identical with those computed using a stability index [16]. Table 2 also shows data from the Cleveland–Armathwaite Dyke [7] and the two studies of the Whin Sill [8, 9]. THe several means signified by asterisks were considered to be most reliable by the authors concerned, and these are plotted in fig. 3.

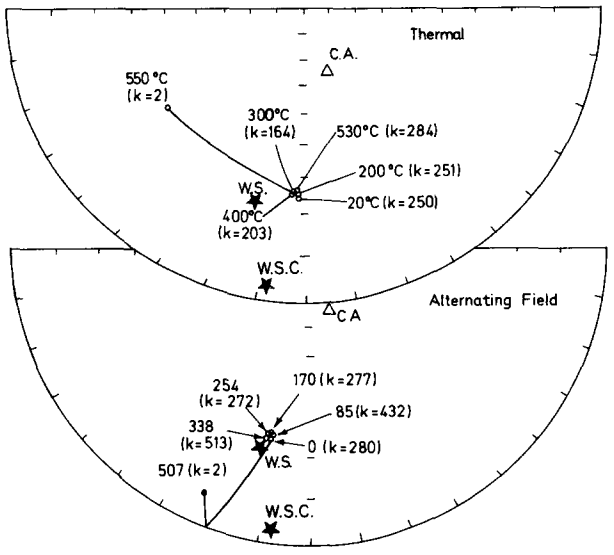


Fig. 3. Mean NRM directions for the Wackerfield Dyke during A.F. and thermal demagnetization. The figure in parenthesis is k, the precision parameter: temperatures are in °C; fields are in oersted. Means are also plotted for the Cleveland-Armathwaite Dyke (CA), and the Whin Sill, as determined by Creer et al. [8], (WSC) and by Storetvedt and Gidskehaug [9] (WS).

We see that the Wackerfield Dyke directions are not the same as those from the Cleveland–Armathwaite Dyke: hence in agreement with the K/Ar age, the dyke is not of Tertiary age.

The mean directions obtained from the 2 studies of the Upper Carboniferous Whin Sill differ by about 17°. Storetvedt and Gidskehaug [9] maintained that their steeper direction, which was confirmed by AF and thermal demagnetization, reflected a true Upper Carboniferous magnetic field, particularly since it agreed with other European data from lavas of similar age. They argued that the shallow mean direction obtained by Creer et al. [8], which represented the initial NRM directions, could be attributed to a stable, secondary, post-Permian magnetization. Here we can note that apparent ages as low as 159 my (mid-Jurassic) have been obtained from the Whin Sill [18], although these were samples which had been very definitely metasomatized.

The overall Wackerfield mean result we report in table 2 is virtually identical with the supposed true Upper Carboniferous field direction of Storetvedt and Gidskehaug [9]. One interpretation of our data is therefore that it supports the idea that apparently unaltered Whin Sill was partially remagnetized at a later date. However other explanations of our data are (a) that it could reflect a structural error, although field evidence suggests this is not the case and (b) the limited extent of the Dyke may mean that secular variation effects have not been averaged out.

Obviously the study we have made on the Wackerfield Dyke allows us to draw only limited conclusions, but it has raised some important questions for future work. We have shown that fresh specimens from the Wackerfield Dyke are isotopically and paleomagnetically of late Carboniferous age and therefore that this dyke is related to the Whin Sill intrusions. A similar study of related dikes in Northern England would therefore amplify mineralogic comparisons between the Paleozoic and Tertiary intrusions [1, 5], and determine the extent of the Upper Carboniferous igneous activity. We have identified various modes of alteration within the Wackerfield Dyke. Paleomagnetic/isotopic studies on other dikes related to the Whin Sill would demonstrate how reset-ages and magnetic-overprinting can together be used as indicators of subsequent episodes of hydro-thermal metasomatism.

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